CPMT: Cross-Plane Masked Prototype-Gated Mean Teacher for CBCT-to-Clinical Record Prediction

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Abstract. Structured clinical prediction from cone-beam computed tomography (CBCT) requires a model to relate three-dimensional anatomy to heterogeneous record fields. This setting is difficult because paired volumes and records are scarce, unpaired scans are more common, and some clinical statements are only partly supported by imaging. We present the Cross-Plane Masked Prototype-Gated Mean Teacher (CPMT), a semi-supervised method for seven-field dental record prediction. CPMT forms a position-aware representation from axial, coronal, and sagittal observations. Training begins with masked EMA self-distillation on all scans. A second stage aligns paired volumes with field, record, and entity targets, while two teacher views admit unlabeled prototype and entity targets only when their predictions are reliable. The trained EMA representation is then matched against a fixed labeled record bank using field, global, and entity similarities. CPMT was trained with 50 paired and 200 unpaired CBCT volumes and obtained a challenge Weighted Score of 0.21 on the 50-case public validation set.

Keywords: Dental CBCT · Semi-supervised learning · Mean Teacher · Clinical record prediction · Retrieval

1 Introduction

Dental CBCT depicts teeth, roots, bone, and treatment materials in three dimensions. Clinical records add information that is not represented in the volume alone, including the chief complaint, history, diagnosis [1], treatment plan, procedure, and advice. MMDental provides paired CBCT volumes and expert seven-field records for studying this relation [2]. Prediction remains difficult because relevant findings may be small, similar structures recur across tooth positions, and several record fields depend on non-imaging evidence [3]. Reviews of dental decision support report the same gap between image recognition and treatment planning [4]. Dental multimodal models also show that a plausible clinical explanation need not be supported by the visible anatomy [5]. Direct free-text

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generation can therefore produce fluent but unsupported records, a problem noted in dental vision–language evaluation.

Semi-supervised 3D multimodal prediction uses paired image–record examples to define the clinical target and unpaired volumes to learn stable anatomical representations. Dental CBCT studies have shown that unlabeled scans can reduce the dependence on dense annotation [6]. Mean Teacher learning supplies slowly varying targets through exponential parameter averaging [7], while reliability tests limit the effect of uncertain pseudo-labels [8]. Cross-plane pretraining treats axial, coronal, and sagittal observations as complementary views of the same volume [9]. Progressive self-supervision and teacher learning have also proved effective for dental CBCT. For clinical prediction, these image objectives must be joined with semantic supervision: dental vision–language models align images with diagnostic concepts [10], and 3D medical vision–language work shows the need to preserve spatial evidence during this alignment [11].

This paper presents CPMT, a semi-supervised method that links cross-plane CBCT evidence to structured clinical records without open-ended generation. A first training stage learns a position-aware volume representation by masked EMA self-distillation on paired and unpaired scans. A second stage aligns paired volumes with field, whole-record, and entity targets. Two weak teacher views then gate prototype and entity pseudo-targets for strongly perturbed unpaired views. The final EMA model retrieves each output field from a fixed labeled record bank using field, global, and entity agreement. This design separates anatomical representation learning from uncertain clinical supervision and keeps every predicted statement within the observed record vocabulary.

2 Method

2.1 Overview and Problem Formulation

Let $\mathcal{D}_l = \{(x_i, r_i)\}_{i=1}^{N_l}$ denote paired CBCT volumes and clinical records, and let $\mathcal{D}_u = \{x_j^u\}_{j=1}^{N_u}$ denote unpaired volumes. Each record $r_i = \{r_i^f\}_{f\in\mathcal{F}}$ contains $F = 7$ fields, where $\mathcal{F} = \{1, \ldots, F\}$. The unpaired volumes supply no record target. For a query x_q, the objective is to estimate $\hat{r}_q = \{\hat{r}_q^f\}_{f\in\mathcal{F}}$ using the learned image representation and the labeled reference bank $\mathcal{B} = \{(x_i, r_i)\}_{i=1}^{N_l}$.

Figure 1 summarizes the method. Stage I matches masked student observations to an EMA teacher on all volumes. Stage II adds paired semantic targets and admits unpaired prototype and entity targets through two agreement gates. The final EMA representation supports the fieldwise prediction in Sec. 2.6.

2.2 Cross-Plane Representation

Let $\mathcal{A} = \{\mathrm{ax}, \mathrm{cor}, \mathrm{sag}\}$ denote the three anatomical planes. For plane $a \in \mathcal{A}$, the operator $\Pi_a(x, p)$ extracts an observation at normalized position $p \in [0, 1]$. Uniform and image-dependent positions provide anatomical coverage and retain distinct structural content. The observation set is

$$\mathcal{O}(x) = \{(I_{a,k}, a, p_{a,k}) : I_{a,k} = \Pi_a(x, p_{a,k}),\ a \in \mathcal{A},\ k = 1, \ldots, K\}. \quad (1)$$

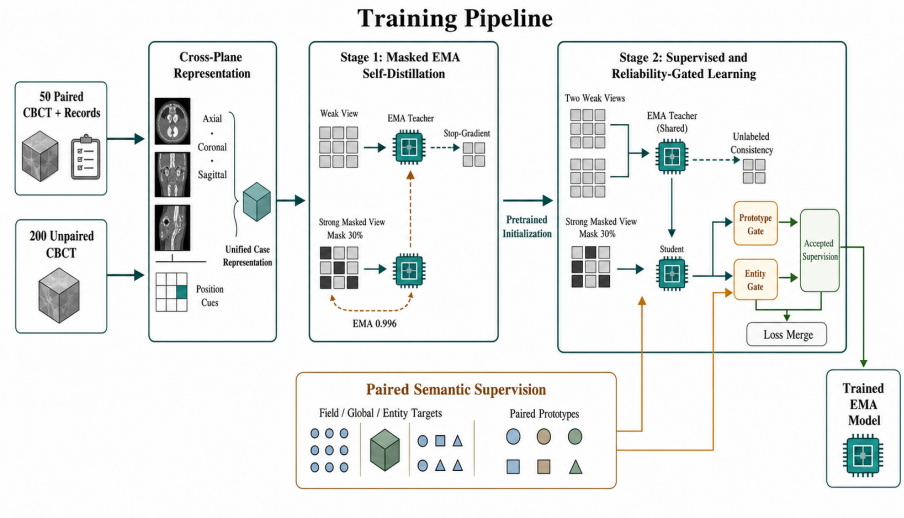


Fig. 1. Training flow of CPMT. Paired and unpaired CBCT volumes share the cross-plane representation and masked EMA self-distillation. Stage II combines paired semantic targets with parallel prototype and entity gates. Only accepted unlabeled targets enter the training loss, and the final EMA model is retained.

Neighboring observations preserve through-plane continuity without dense voxel prediction [12].

Let ϕ_θ map an observation to a visual vector, e_a encode its plane, and $\rho(p) = [p, \sin(2\pi p), \cos(2\pi p)]$ encode its position. A position-conditioned observation and the unified volume representation are defined by

$$v_{a,k} = \phi_\theta(I_{a,k}) + e_a + \psi_\theta(\rho(p_{a,k}))\,, \tag{2}$$

$$h_\theta(x) = \mathrm{Agg}_\theta(\{v_{a,k}\}_{a,k})\,. \tag{3}$$

Here Agg_θ is a permutation-invariant map that relates distant observations [13]. The formulation is independent of a specific implementation and bounds full-volume computation through a finite observation set.

The task map returns normalized field embeddings, a normalized whole-record embedding, and entity probabilities:

$$\mathcal{E}_\theta(x) = \Big(\{z^f_\theta(x)\}_{f\in\mathcal{F}}, z^r_\theta(x), p^e_\theta(x)\Big)\,, \quad \|z^f_\theta\|_2 = \|z^r_\theta\|_2 = 1. \tag{4}$$

2.3 Stage I: Masked EMA Self-Distillation

Stage I uses $\mathcal{D}_l \cup \mathcal{D}_u$ without record targets. Let a_w and a_s be weak and strong transformations, $\mathcal{M}$ an observation mask, and q_θ the case map in Eqs. (1)–(3). Student and teacher representations are

$$u_S = \mathrm{norm}(q_{\theta_S}(\mathcal{M}(a_s(x))))\,, \qquad u_T = \mathrm{norm}(q_{\theta_T}(a_w(x)))\,, \tag{5}$$

with analogous plane-specific representations u_S^a and u_T^a. For normalized vectors, define the stopped-gradient cosine distance $d_c(u, v) = 1 - u^\top \mathrm{sg}(v)$. The distillation objective is

$$\mathcal{L}_{\mathrm{ssl}} = d_c(u_S, u_T) + \frac{\beta_p}{|\mathcal{A}|} \sum_{a \in \mathcal{A}} d_c(u_S^a, u_T^a) + \beta_v \mathcal{L}_{\mathrm{var}}. \tag{6}$$

The variance term acts across a minibatch $\mathcal{Q}$ and prevents a constant representation:

$$\mathcal{L}_{\mathrm{var}} = \frac{1}{d} \sum_{j=1}^{d} \left[\max\left(0, \gamma - \sqrt{\mathrm{Var}_{x \in \mathcal{Q}}(u_{S,j}(x)) + \epsilon}\right) \right]^2. \tag{7}$$

This penalty directly excludes collapsed representations and follows variance-regularized semi-supervised learning.

The teacher is updated only by an exponential moving average [7]:

$$\theta_T \leftarrow \alpha \theta_T + (1 - \alpha)\theta_S. \tag{8}$$

Thus Stage I learns invariance to perturbation and missing observations before record supervision.

2.4 Stage II: Supervised Semantic Alignment

Each paired record is mapped to deterministic semantic targets. For a text string s, let $\mathcal{G}(s)$ be its character n-grams, $H(g)$ a fixed hash, and $\sigma(g) \in \{-1, 1\}$ a fixed sign. The normalized text target is

$$t(s) = \mathrm{norm}\left(\sum_{g \in \mathcal{G}(s)} \sigma(g) e_{H(g) \bmod D} \right), \tag{9}$$

where e_j is the jth coordinate vector in $\mathbb{R}^D$. It yields field targets t_i^f and a record target t_i^r without generated-text drift. Clinical terms are mapped separately to a multi-hot entity vector y_i^e.

Let z_i^f, z_i^r, and p_i^e be the outputs in Eq. (4). Field alignment uses cosine distance,

$$\mathcal{L}_{\mathrm{field}} = \frac{1}{N_l F} \sum_{i=1}^{N_l} \sum_{f \in \mathcal{F}} \left(1 - (z_i^f)^\top t_i^f\right). \tag{10}$$

Whole-record alignment uses a symmetric contrastive objective. With

$$\ell_{\mathrm{NCE}}(a_i, b_i) = -\log \frac{\exp(a_i^\top b_i / \tau)}{\sum_{j=1}^{N_l} \exp(a_i^\top b_j / \tau)}, \tag{11}$$

the global and entity losses are

$$\mathcal{L}_{\text{global}} = \frac{1}{2N_l} \sum_{i=1}^{N_l} \left[\ell_{\text{NCE}}(z_i^r, t_i^r) + \ell_{\text{NCE}}(t_i^r, z_i^r) \right], \tag{12}$$

$$\mathcal{L}_{\text{entity}} = \frac{1}{N_l} \sum_{i=1}^{N_l} \text{BCE}_w(p_i^e, y_i^e). \tag{13}$$

Weighted binary cross-entropy accounts for sparse entities; the contrastive term follows limited-pair 3D image–text adaptation. The paired objective is

$$\mathcal{L}_{\text{sup}} = \mathcal{L}_{\text{field}} + \lambda_g \mathcal{L}_{\text{global}} + \lambda_e \mathcal{L}_{\text{entity}}. \tag{14}$$

2.5 Stage II: Reliability-Gated Teacher Learning

For an unpaired volume, the EMA teacher receives two independent weak views, indexed by $v \in \{1, 2\}$, and the student receives a strongly transformed, masked view. The averaged teacher embeddings $\bar{z}_T^f = \text{norm}(z_{T,1}^f + z_{T,2}^f)$ and $\bar{z}_T^r = \text{norm}(z_{T,1}^r + z_{T,2}^r)$ define the continuous consistency loss

$$\mathcal{L}_{\text{cons}} = \frac{1}{F} \sum_{f \in \mathcal{F}} d_c(z_S^f, \bar{z}_T^f) + d_c(z_S^r, \bar{z}_T^r). \tag{15}$$

Discrete clinical targets require separate prototype and entity gates [8].

For field f, the labeled targets $\{t_i^f\}_{i=1}^{N_l}$ act as semantic prototypes. For teacher view v, define

$$s_{v,i}^f = (z_{T,v}^f)^\top t_i^f, \qquad k_v^f = \arg\max_i s_{v,i}^f, \qquad \bar{s}_i^f = \tfrac{1}{2}(s_{1,i}^f + s_{2,i}^f). \tag{16}$$

Let $\bar{s}_{(1)}^f$ and $\bar{s}_{(2)}^f$ be the two largest averaged similarities. The prototype gate is

$$g_f^p = \mathbf{1}[k_1^f = k_2^f]\, \mathbf{1}[\bar{s}_{(1)}^f - \bar{s}_{(2)}^f \geq \delta], \qquad \mathcal{L}_{\text{proto}} = \frac{\sum_f g_f^p d_c(z_S^f, t_{k_1^f}^f)}{\sum_f g_f^p + \epsilon}. \tag{17}$$

View agreement rejects unstable assignments, and the margin rejects ambiguous neighbors [14].

For entity m, let $\bar{p}_m = \frac{1}{2}(p_{T,1,m}^e + p_{T,2,m}^e)$. Confidence and stability define

$$g_m^e = \mathbf{1}[\max(\bar{p}_m, 1 - \bar{p}_m) \geq \tau_c]\, \mathbf{1}[|p_{T,1,m}^e - p_{T,2,m}^e| \leq \tau_s], \tag{18}$$

$$\tilde{y}_m^e = \mathbf{1}[\bar{p}_m \geq \tfrac{1}{2}], \qquad \mathcal{L}_{\text{pe}} = \frac{\sum_m g_m^e \, \text{BCE}(p_{S,m}^e, \tilde{y}_m^e)}{\sum_m g_m^e + \epsilon}. \tag{19}$$

The two gates run in parallel and merge only at the unlabeled objective:

$$\mathcal{L}_u = \lambda_c \mathcal{L}_{\text{cons}} + \lambda_p \mathcal{L}_{\text{proto}} + \lambda_{pe} \mathcal{L}_{\text{pe}}, \tag{20}$$

$$\mathcal{L}_{\text{stage2}} = \mathcal{L}_{\text{sup}} + \lambda(t) \mathcal{L}_u. \tag{21}$$

The ramp $\lambda(t)$ delays unlabeled supervision until the semantic space is stable [15]. Equation (8) continues through Stage II, and the final EMA state is retained.

2.6 Fieldwise Clinical Retrieval

The final EMA model maps a query x_q to z_q^f, z_q^r, and p_q^e. For labeled reference i, field and whole-record similarities are $s_f(q,i) = (z_q^f)^\top t_i^f$ and $s_r(q,i) = (z_q^r)^\top t_i^r$. Let $\mathcal{E}_f$ be the entity types relevant to field f. Their soft intersection-over-union is

$$s_e^f(q,i) = \frac{\sum_{m\in\mathcal{E}_f} \min(p_{q,m}^e, y_{i,m}^e)}{\sum_{m\in\mathcal{E}_f} \max(p_{q,m}^e, y_{i,m}^e) + \epsilon}. \tag{22}$$

Global and entity evidence follows region-grounded report prediction. The global anchor is

$$i^* = \arg\max_i s_r(q,i). \tag{23}$$

The score for field f is

$$S_f(q,i) = w_f s_f(q,i) + w_r s_r(q,i) + w_e s_e^f(q,i) + w_a \mathbf{1}[i = i^*]. \tag{24}$$

The predicted text is selected only from nonempty labeled values:

$$\hat{i}_f = \arg\max_{i: r_i^f \neq \varnothing} S_f(q,i), \qquad \hat{r}_q^f = r_{\hat{i}_f}^f. \tag{25}$$

Fieldwise selection weights evidence by clinical field; the fixed bank prevents unseen output text.

3 Experiments

3.1 Dataset and Evaluation

The MMDental Task 3 data contained 50 labeled CBCT volumes paired with seven-field records, 200 unlabeled volumes, and 50 public validation volumes with hidden records [2]. Final training used all labeled and unlabeled volumes. The public cases were evaluated on Codabench [16].

The challenge Weighted Score combines lexical and structured clinical agreement. BLEU measures modified precision for n-grams of orders one to four [17]. METEOR and ROUGE-L measure token alignment and longest common subsequence overlap. CIDEr weights consensus n-grams by their corpus frequency [18]. The structured measures report precision, recall, and F1 for tooth notation, diagnosis codes, treatment actions, and medication, together with diagnosis matching, treatment-plan recall, and record completion.

3.2 Implementation Details

Preprocessing. Volumes were permuted and flipped to the nearest axis-aligned RAS+ layout without spacing resampling. Finite intensities were clipped case-wise at the 0.5th/99.5th percentiles and scaled to $[0,1]$. A coarse mask using the larger of 0.03 and P_{20} defined a bounding box with an 8-voxel margin. Per axis,

Table 1. Training protocol and recorded evaluation environment.

Item	Setting
Training data	50 paired and 200 unpaired CBCT volumes
Software	Python 3.10; PyTorch 2.1.2+cu118; torchvision 0.16.2+cu118
Representation / parameters	3 planes × 8 locations / 13,089,726
Stage 1	12 epochs; batch 4; mask ratio 0.30; EMA 0.996
Stage 2	24 epochs; labeled/unpaired batches 2/2
Optimization	AdamW; 3×10^{-4} / 3×10^{-5}; cosine schedule
Entity vocabulary / updates	100 labels / 2400 total updates
Retrieval weights	$(w_f, w_r, w_e, w_a) = (0.65, 0.20, 0.15, 0.05)$
Evaluation environment	Ubuntu 20.04; RTX 3090 24 GB; Xeon 6248R; 28 GB RAM

four uniform locations and four of 32 candidates ranked by intensity standard deviation, mean absolute in-plane gradients, and 0.15× foreground occupancy were retained with separation. The RAS+ axes yielded sagittal-, coronal-, and axial-normal views.

Visual encoder. Each location used the center and adjacent normal slices (offsets $-1, 0, +1$, boundary-clamped) as a three-channel 2.5D observation. It was resized to 160×160 and stored as uint8; training restored $[0, 1]$ before ImageNet normalization. A classifier-free ResNet-18 [19] gave a 512-D feature; a linear layer and LayerNorm produced a 192-D token. The run requested torchvision ImageNet-1K initialization with random fallback. Only residual layers 3–4 were trainable; the stem and layers 1–2 were frozen.

Cross-plane aggregation. Each of the 24 tokens was summed with a plane embedding and a $3 \to 192 \to 192$ GELU MLP of $[p, \sin(2\pi p), \cos(2\pi p)]$. A learned case token was prepended before a two-layer pre-norm Transformer with four heads, 768-D feed-forward layers, GELU, and dropout 0.1. Its case output drove seven 256-D field heads, one 256-D record head (all L_2-normalized), a 128-D distillation head, and a 100-logit entity head. Within-plane token means used a shared 128-D distillation head.

Training. Weak views used mild intensity scaling and shifts; strong views added gamma/contrast, Gaussian noise ($\sigma = 0.035$), flips, and Cutout. A 30% token mask kept at least one observation per plane. AdamW used weight decay 0.01, cosine scheduling, mixed precision, and gradient clipping at 5.0. With $D = 256$, the loss weights were $(\beta_p, \beta_v, \gamma) = (0.50, 0.05, 0.25)$, $(\lambda_g, \lambda_e) = (0.35, 0.80)$, and $(\lambda_c, \lambda_p, \lambda_{pe}) = (0.5, 0.7, 0.3)$. The gates used $(\delta, \tau_c, \tau_s) = (0.06, 0.8, 0.15)$; ramp-up ended at 35% of Stage II. Table 1 summarizes the remaining protocol.

4 Results and Discussion

4.1 Challenge Results

CPMT returned a complete seven-field record for all 50 public validation volumes and obtained a Weighted Score of 0.21. Table 2 reports the challenge re-

Table 2. Challenge results on the 50-case public validation set. P, R, and F1 denote precision, recall, and F1 score.

Metric	Score
Weighted Score	0.21
BLEU-1 / BLEU-2 / BLEU-3 / BLEU-4	0.17 / 0.06 / 0.03 / 0.02
METEOR / ROUGE-L / CIDEr	0.20 / 0.20 / 0.05
Prediction / reference completion	1.00 / 0.90
Tooth notation P/R/F1	0.11 / 0.23 / 0.10
Diagnosis code P/R/F1	0.07 / 0.09 / 0.07
Treatment action P/R/F1	0.28 / 0.45 / 0.31
Medication P/R/F1	0.29 / 0.28 / 0.28
Diagnosis exact / partial	0.04 / 0.09
Treatment-plan recall	0.51

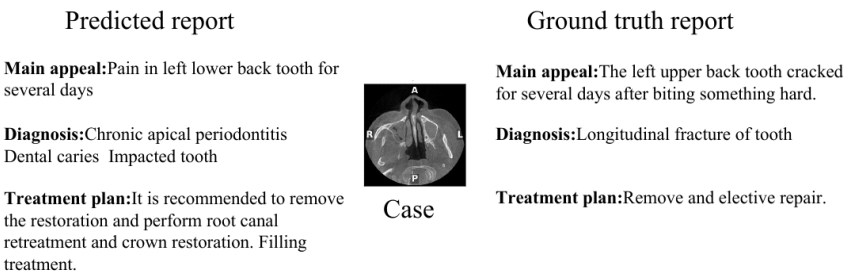


Fig. 2. Internal validation example with an axial CBCT slice and predicted and ground truth text for three record fields.

sults. Lexical agreement was modest: BLEU-1 was 0.17, while BLEU-4 was 0.02, and both METEOR and ROUGE-L were 0.20. These values indicate overlap in clinical terms but limited agreement in longer phrases.

Treatment was the strongest structured category. Treatment-action F1 was 0.31, and treatment-plan recall was 0.51. The F1 scores were 0.10 for tooth notation and 0.07 for diagnosis codes, showing weaker location and diagnosis agreement.

Figure 2 shows an internal labeled validation case, not an official public case. The complete prediction confuses upper and lower dentition, diagnosis, and treatment, reflecting the gap between broad clinical coverage and exact agreement. This agrees with the low tooth and diagnosis scores in Table 2.

4.2 Limitations

Sparse cross-plane observations may miss small or tooth-specific findings, as suggested by Fig. 2. Denser spatial evidence may reduce this error at greater computational cost. Some clinical fields may have no imaging correlate, including complaint, history, and recommendations, while a fixed bank cannot express

concepts absent from the 50 paired records. Independent field retrieval can also combine incompatible values. Finally, teacher-view agreement cannot exclude shared systematic error. Spatially grounded evidence, controlled ablations, and record-level constraints are therefore required.

5 Conclusion

CPMT learns cross-plane CBCT representations by masked EMA distillation, aligns paired volumes with record semantics, and gates unpaired prototype and entity targets. Its final EMA representation supports constrained fieldwise retrieval. With 50 paired and 200 unpaired volumes, CPMT achieved a challenge Weighted Score of 0.21 with complete output coverage. Remaining errors in tooth location and diagnosis motivate stronger spatial and record-level constraints.

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